

SELVAKUMAR SADURSHAN

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Summary

Aspiring Machine Learning Engineer passionate about applying Data Science and AI to solve real-world problems. Skilled in Python, Machine Learning, Deep Learning, data analysis, SQL, REST APIs, and Git. Familiar with Generative AI, LLMs, Prompt Engineering, and RAG. Experienced with Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch. Knowledgeable in AWS cloud services and continuously learning MLOps, LLMOps, and Agentic AI to develop scalable AI solutions

Professional Experience

Intern - Data analyst, HSBC 08/2025 – 02/2026

- Performed data extraction, cleaning, and validation to ensure data accuracy and compliance with the Personal Data Protection Act
- Developed Power BI and Excel dashboards to analyze department performance and work quality distribution
- Automated email processes using Excel Macros to improve efficiency
- Utilized Python (PyCharm) for data cleaning, data matching, and generating name combinations for accurate customer identification
- Created scripts to identify customers with account transfers and detect ineligible customers
- Supported data-driven decision-making through reports, visualizations, and collaboration with teams

Student Counselor, PrimeleedManagement 06/2022 – 10/2022

- Served as the main point of contact for UK clients and universities, ensuring clear communication and strong relationships.
- Guided students on course selection, university admissions, and application processes.
- Worked with the team to meet monthly targets for student enrolment and client satisfaction.

Course-Consultant Officer 08/2021 – 02/2022

- Conducted workshops to explain e-club courses in an interactive way.
- Helped students gain practical, hands-on skills through these sessions.
- Promoted e-club courses on social media to increase awareness.
- Created engaging content and regular updates to reach more students.
- Encouraged student participation and enrolment in e-club programmes.

Education

Bachelor's Degree in Data-Science, SLIIT 2022 – Present

Certificate level of BCS Higher Education Qualification (HEQ), Live College
BCS, The Chartered Institute for IT : 1st Stage Completed

Diploma In IOT, Esoft Metro Campus
Basics Of IOT

Skills

Programming Languages

R, C, Java, Python, C++, kotlin

Mobile Development

Android Application Development

Data-warehousing

AI Libraries

Tensor-flow, Skilearn, kears, Pytorch

Web Development

MERN Stack, Spring-Boot

UI/UX Design

UI Designing, Figma, Draw.io

Tools

Power-BI, Trello, Kanban board

Certifications

- Introduction to Party Rock : AWS Educate
- Prompt Engineering : Greeting Learning
- Introduction to Generative Ai : AWS Educate
- IBM data-Analyst : Udemey
- IBM Machine Learning With Python : Coursera

Project

Anomaly Detection on turbofan engine of aircraft

- Developed an Explainable Health Monitoring & Anomaly Detection System for turbofan engines using the NASA C-MAPSS dataset
- Implemented context-aware anomaly detection to distinguish operational changes from actual faults
- Built Digital Twin models using LSTM networks for expected sensor behavior prediction
- Applied Isolation Forest and LSTM Autoencoder models for anomaly detection
- Generated residual-based anomaly analysis using predicted vs actual sensor values
- Performed feature engineering, preprocessing, and time-series sequence generation using Python
- Used Python libraries including Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch
- Designed explainable AI outputs with root-cause reasoning and anomaly interpretation
- Developed data visualization and monitoring dashboards for anomaly insights
- Worked with large-scale sensor data processing and machine learning workflows

Data cleaning and visualization Using Python

- Conducted Exploratory Data Analysis (EDA) on Netflix datasets using Python to uncover trends in content, genres, and ratings.
- Performed data cleaning and transformation using Pandas.
- Built visualizations with Matplotlib and Seaborn to present key insights and support effective data storytelling.

Climatic Policy Analysis System

- Developed an NLP-based system to analyze and compare climate policy documents
- Implemented document upload functionality supporting PDF, Word, and text formats
- Performed text preprocessing and cleaning (noise removal, normalization, input sanitization)
- Designed a document analyzer using keyword extraction and Named Entity Recognition (NER)
- Built a policy comparison module to identify overlaps and unique elements between policies
- Generated similarity scores to evaluate policy alignment
- Developed a recommendation engine to suggest improvements based on analysis results
- Applied machine learning and NLP techniques to extract insights from unstructured data

IOT_Parking_System_And_ML

- Developed a smart parking system using IoT devices (Ultrasonic sensors, IR sensors, Arduino) to collect real-time parking data
- Built machine learning models to analyze parking patterns and predict space availability
- Implemented anomaly detection to identify unusual parking behavior
- Applied clustering techniques to group similar parking usage patterns
- Performed data cleaning and preprocessing for accurate analysis
- Created visualizations to show parking trends and insights
- Developed backend (Node.js) and frontend for system operation and monitoring
- Organized the project using a structured and scalable architecture

RAG_Document_analyser

- Developed a Retrieval-Augmented Generation (RAG) based AI Document Assistant for intelligent PDF querying and contextual response generation
- Implemented PDF text extraction, preprocessing, and semantic text chunking using Python and LangChain
- Generated vector embeddings using SentenceTransformers and stored them in a FAISS vector database
- Integrated transformer-based Large Language Models (LLMs) for context-aware answer generation
- Built REST API endpoints for document upload and AI-powered querying using FastAPI
- Applied NLP, semantic similarity search, vector databases, and Generative AI concepts in the project
- Technologies used: Python, FastAPI, LangChain, Hugging Face Transformers, SentenceTransformers, FAISS, and PyTorch